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Maritime & Coastguard Agency

Guidance

MGN 553 Amendment 1 Annex 1

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Annex 1 Recommended standard for the conditions of service stations and conduct of servicing for non-solas inflatable liferafts.

1. General

1.1 Scope

This standard aims to define best practice for the service of small craft inflatable liferafts. It is intended to be compatible with all small craft inflatable liferafts – past, present and future; however, it has been drafted with particular reference to liferafts compliant with ISO 9650 Parts 1, 2 and 3 and/or the International Sailing Federation (ISAF) Offshore Special Regulations, Appendix A Part II (and Part I).

1.2 Definitions

• Manufacturer

The manufacturer is any natural or legal person who is responsible for designing or manufacturing a product and places it on the market under his own name or trademark. So, if the product is marked under another person's name or trademark, this person will be considered as the manufacturer.

• Service Stations

Companies offering small craft inflatable liferaft servicing under this standard.

• Small craft

Any water borne vessel <24 metres in length.

1.3 Responsibilities

In order to ensure that the service of inflatable liferafts is effectively conducted to provide a reliable survival craft in an emergency, Manufacturers and liferaft owners have parallel and overlapping responsibilities; these include, but are not limited to, the following:

1.3.1 **Manufacturers** are responsible for:

1.3.1.1 ensuring that their liferafts can be adequately serviced in accordance with this standard or with any additional requirements necessary for that particular product;

1.3.1.2 accrediting a sufficient number of service stations;

1.3.1.3 ensuring that each service station accredited by them for service and repair of their liferafts has qualified persons whom they have adequately trained and certificated to perform such work; the training procedure should also ensure that servicing personnel are made aware of changes and new techniques.

1.3.1.4 making available to accredited service stations:

changes to any Manufacturers service manuals, service bulletins and instructions;

the correct materials and replacement parts;

up to date copies of this standard.

1.3.1.5 keeping maritime Administrations fully informed of any casualties known to them and involving their liferafts and also of any failures of liferafts;

1.3.1.6 informing liferaft owners whenever possible of any deficiency or danger known to them and related to the use of their liferafts and taking whatever remedial measures they deem necessary.

1.3.2 **Liferaft owners** are responsible for:

1.3.2.1 . ensuring that liferafts are serviced at the appropriate intervals at an accredited service station. Liferaft owners should check that station can complete the servicing of raft by checking for certification from the product manufacturer, if in doubt call the product manufacturer.

1.3.2.2 Whenever practicable owner should be in attendance during service inflation tests to see the raft in its inflated condition.

2. Requirements for Service Stations

2.1 Service stations should demonstrate competence to service and repack liferafts, maintain an adequate facility and use only qualified persons who have been adequately trained and certificated. In particular they should:

2.1.1 only conducting servicing on liferaft for which they are accredited by the manufacturer and for which they hold the necessary training, information, equipment, spares etc;

2.1.2 only carry out the service of inflatable liferafts in fully enclosed spaces. There should be ample room for the number of inflatable liferafts expected to be serviced at any one time; the ceiling should be sufficiently high to allow the largest liferafts to be serviced to be turned over when inflated, or an equally efficient means to facilitate inspection of bottom seams should be provided;

2.1.3 provide a service space floor that is sufficiently clean and smooth to ensure that no damage will occur to the liferaft fabric;

2.1.4 provide a well-lit service space that the direct rays of sunlight do not enter;

2.1.5 ensure the temperature and, when necessary, the relative humidity in the service space is sufficiently controlled to ensure that service and repairs can be effectively carried out;

2.1.6 ensure the service space is efficiently ventilated, but be free from draughts;

2.1.7 ensure separate areas or rooms are provided for:

- 2.1.7.1 liferafts awaiting service, repair or delivery;
- 2.1.7.2 the repair of glass-fibre containers and the refurbishment of compressed gas cylinders;
- 2.1.7.3 materials and spare parts;
- 2.1.7.4 unused and expired pyrotechnics;
- 2.1.7.5 administrative purposes;

2.1.8 provide means, in the liferaft storage space, to ensure that liferafts in containers or valises are neither stored on top of each other in more than two tiers unless supported by shelving nor that they are subjected to excessive loads;

2.1.9 ensure that spare and obsolete pyrotechnics or other hazardous materials such as Li-ion batteries from position indicating lights are stored in separate, safe and secure magazines well away from the service and storage spaces and that they meet any local authority requirements including the disposal of such items. All such items should be stored in accordance with the equipment manufacturer’s instructions;

2.1.10 make available sufficient facilities, tools and equipment for the service of liferafts in accordance with the requirements of the

Manufacturer, including:

2.1.10.1 suitable and accurate manometers or pressure gauges, thermometers and barometers which can be easily read;

2.1.10.2 one or more air pumps for inflating and deflating liferafts, together with a means of cleaning and drying the air and including the necessary high-pressure hoses and adapters;

2.1.10.3 a scale for weighing inflation gas cylinders with sufficient accuracy;

2.1.10.4 sufficient dry compressed gas for blowing through the inlet system of the liferafts and for liferaft inflation;

2.1.10.5 all measurement tools and equipment should be calibrated and certificated as recommended by the Manufacturers;

2.1.11 establish procedures to ensure that each gas cylinder is properly filled and gastight before fitting to a liferaft;

2.1.12 ensure sufficient materials and accessories are available for repairing liferafts, together with replacements of the emergency equipment to the satisfaction of the Manufacturer;

2.1.13 ensure service and repair work is carried out only by qualified persons who have been adequately trained and certificated by the liferaft Manufacturer. Service stations should have procedures to ensure that service personnel are made aware of changes and new techniques;

2.1.14 make arrangements for the Manufacturer to make available to the service station:

2.1.14.1 changes to service manuals, service bulletins and instructions;

2.1.14.2 the correct replacement materials and replacement parts;

2.1.14.3 training for service technicians;

2.1.15 ensure any persons conducting work on behalf of the service station, including the use of sub-contractors for the servicing of gas cylinder, operate to the same requirements and standards. This is likely to involve Quality Assurance activities.

3. Information to Liferaft Owners

Manufacturers should provide information for liferaft owners regarding service facilities for inflatable liferafts; where possible the manufacturer should include the contact details with all the relevant information in the operating and maintenance manual;

4. Service of inflatable liferafts

The following tests and procedures should be carried out, except where noted otherwise, at every service of an inflatable liferaft. Inflatable liferaft service should be carried out in accordance with the appropriate Manufacturer’s service manual. Necessary procedures should include, but not be limited to, the following:

4.1 inspection of the container / valise for damage;

4.2 inspection of the folded liferaft and the interior of the container for signs of dampness;

4.3 inspection of the valise packed folded liferaft and interior of valise for signs of damage;

4.4 a gas inflation (GI) test should be carried out at intervals as per Appendix 3. When undertaking a gas inflation test, special attention should be paid to the effectiveness of the relief valves. The folded liferaft should be removed from its container before activating the fitted gas inflation system. After gas inflation has been initiated, sufficient time should be allowed to enable the pressure in the buoyancy tubes to become stabilized and the solid particles of the inflation gas to evaporate. After this period the buoyancy tubes should, if necessary, be topped up with air, and the liferaft subjected to a pressure holding test; each liferaft should be subjected to the necessary additional pressure (NAP) test as described in Appendix 1 at intervals as per Appendix 3 at yearly intervals after the tenth year of the liferaft`s life unless earlier service is deemed necessary as a result of visual inspection. After allowing sufficient time for the liferaft to regain fabric tension at working pressure, the liferaft shall be subjected to a pressure holding test over a period of not less than one hour during which the pressure drop should not exceed 5% of the working pressure;

4.5 when a Necessary Additional Pressure (NAP) or Gas Inflation (GI) test is not required, a working pressure (WP) test should be carried out at intervals as per Appendix 3, by inflation of the liferaft with dry compressed air, after removing it from the container shell or valise and from its retaining straps, if fitted, to at least the working pressure, or to the pressure required by the Manufacturer’s service manual if higher. The liferaft should be subjected to a pressure holding test over a period of not less than one hour, during which the pressure drop will not exceed 5% of the working pressure;

4.6 while inflated, the liferaft should be subjected to a thorough inspection inside and out in accordance with the Manufacturer’s instructions;

4.7 the floor, if of an inflatable type, should be inflated, checked for broken reeds and tested in accordance with the Manufacturer’s instructions;

4.8 the floor seam strength (FSS), that is the strength of the seams between floor and buoyancy tube, should be checked for slippage or edge lifting at intervals as per Appendix 3. With the buoyancy tubes supported by a system which leaves the floor seams unsupported, at a suitable height above the service floor as shown in Appendix 3, a person weighing not less than 75kg should walk/crawl round the perimeter of the floor for the entire circumference and the floor seams should be checked again. Manufacturers may substitute this test with another test which will determine the integrity of the floor seam until the next inspection is due.

4.9 after deflation, any arch roots and/or inflatable boarding system should be checked in accordance with the Manufacturer’s instructions;

4.10 all items of equipment should be checked to ensure that they are in good condition and that dated items are replaced at the time of service if their expiry dates occur before the date of the next service as per Appendix 3;

4.11 a check should be made to ensure that the liferaft and the atmosphere are dry when the liferaft is being repacked;

4.12 the required markings should be updated and checked with particular attention paid to lists of fitted or enclosed life-saving equipment and their service expiry dates and the markings an compressed gas cylinder;

4.13 a record of service, recording all test, inspections, and including digital photographs of any defects, etc. and actions undertaken, should be maintained for at least 5 years after the date of service. Such records, should be available to the Manufacturer;

4.14 Records should be kept on all liferafts serviced, indicating, in particular, defects found, repairs carried out and units condemned and withdrawn from service. A copy of such records should be passed to the owner and Manufacturer;

4.15 Cylinder refurbishment should be carried out as per manufacturer instructions. The servicing of compressed gas cylinder should be carried out at intervals as per Appendix 3. All compressed gas cylinders should be weighed and checked against the gross mass which has been marked on the filled cylinder. To allow for difference of scales when check-weighing, a tolerance of +/-14 g is permitted. No gas cylinder should be fitted to a liferaft unless it has passed one of the following two tests:

- For gases other than CO_2. Storage for a period of at least 30 days after filling. Weighing should take place before and after storage using the same scales. There shall be no loss of weight.
- For CO_2 gas only. As above or the leak test as specified in Appendix 2

5. Service intervals

Small craft inflatable liferaft service intervals should be as Appendix 3 except when the Manufacturer recommends lesser intervals in which case the Manufacturer's Recommendations should be used.

Appendix 1

Necessary additional pressure (NAP) test

1. Plug the pressure relief valves.

• Gradually raise the pressure to the lesser of 2.0 times the working pressure or impose a tensile load on the inflatable tube fabric of at least 20% of the minimum required tensile strength.

• After 5 minutes, there should be no seam slippage, cracking, or other defects (IMO resolution A.521 (13)), part 1, paragraph 5.18.4.1), or significant pressure drop. If cracking in the buoyancy tubes is audible, the liferaft should be condemned; if no cracking is heard, the pressure in all buoyancy chambers should be reduced simultaneously by removing the plugs from the pressure relief valves.

2. Liferaft Manufacturers should include tables in their service manuals of exact NAP test pressures corresponding to their particular tube sizes and fabric tensile strength requirements, calculated according to the equation:

$$pressure\left(kg/cm^2\right)=\frac{2\times tensile\ strength\left(kg\ per\ 5cm\right)}{25\times diameter\left(cm\right)}$$

Appendix 2

Cylinder leak test – CO_2 only

1. Required Materials

i) Polythene bags of a suitable size to fit over the head of the cylinder, e.g. a. for

   a. 125mm diameter cylinder the bag size is approximately 230mm open width x 300mm length

   b. for a 100mm diameter cylinder the bag size is approximately 165mm open width x 300mm length

   c. for a 90mm diameter cylinder the bag size is approximately 150mm open width x 300mm length

ii) Elastic bands of a suitable size.

iii) A measuring glass, capacity 25 ml.

2. Test solution

i) The test solution should be as recommended by the Manufacturer used to indicate small amounts of CO_2 gas.

ii) The solution should be stored as recommended by the Manufacturer.

3. Test Method

i) Lay the cylinder to be tested on its side in a rack such that the valve end is protruding. Make sure the valve and shoulder of the cylinder are free from dust and other contaminants by carefully wiping with a clean, dry cloth. Remove the dust cap and clean the valve. Replace the cap loosely.

ii) Using the measuring glass transfer 25ml of the test solution into a polythene bag.

iii) Pass the open end of the bag over the valve head and seal this to the cylinder body using one or more elastic bands. Make sure there are no air gaps in the seal.

iv) The polythene bag should hang 200mm off the valve end of the cylinder with the test solution in one corner.

v) Maintain the test for a period of not less than one hour.

vi) After the test period shake the solution gently and the observations detailed in paragraph 4.

A control sample is necessary to detect any contamination. The sample is made by pouring 25ml of test solution into a bag which is not fitted to a cylinder but is sealed at the open end with adhesive tape to exclude atmospheric contamination. This bag should be placed on the rack in the vicinity of the cylinders being tested.

4. Tests Observations

i) If no colour change is observed there is no leak of gas from the cylinder.

ii) A leak of CO_2 from the cylinder will cause the pink colour of the test solution to fade. The test solution will become clear as water.

iii) The control sample should not change colour during the test. If a colour change takes place, this indicates that the atmosphere in the test area is contaminated with carbon dioxide and tests carried out together with this control sample are invalid. Tests should be repeated after corrective action has been taken on the atmosphere.

Appendix 3

Small craft inflatable liferaft service intervals should be as follows except when the Manufacturer recommends shorter intervals in which case the Manufacturer's recommendations shall be used.

Valise packed liferafts service intervals are less than those for canister packed liferafts owing to their being more vulnerable to damage.

Frequency of tests for:-

- working pressure (WP)
- gas inflation (GI)
- floor seam strength (FSS)
- necessary additional pressure (NAP)
- cylinder refurbishment (CR)

Canister packed liferafts

Service intervals	Required Test(s)
End of third year	WP
End of fifth year *	GI
End of year six and annually onwards when no other test undertaken	WP

Canister packed liferafts

Service intervals	Required Test(s)
End of tenth year	GI + FSS + CR
End of thirteenth year	NAP + FSS
End of fifteenth year	GI + NAP + FSS
End of eighteenth year	NAP + FSS
End of twentieth year	GI + NAP + FSS + CR
Twenty-second year and annually onwards	GI + NAP + FSS + CR**

*may be at end of the sixth year if so recommended by the Manufacturer

**Unless the cylinder is replaced and then at five year intervals

Valise packed liferafts

Service intervals	Required Test(s)
End of each year when no other test undertaken	WP
End of fifth year *	GI
End of tenth year	GI t + FSS + CR
End of thirteenth year	NAP + FSS
End of fifteenth year	GI + NAP + FSS
End of eighteenth year	NAP + FSS
End of twentieth year	GI + NAP + FSS + CR
Twenty-second year and annually onwards	GI + NAP + FSS + CR**

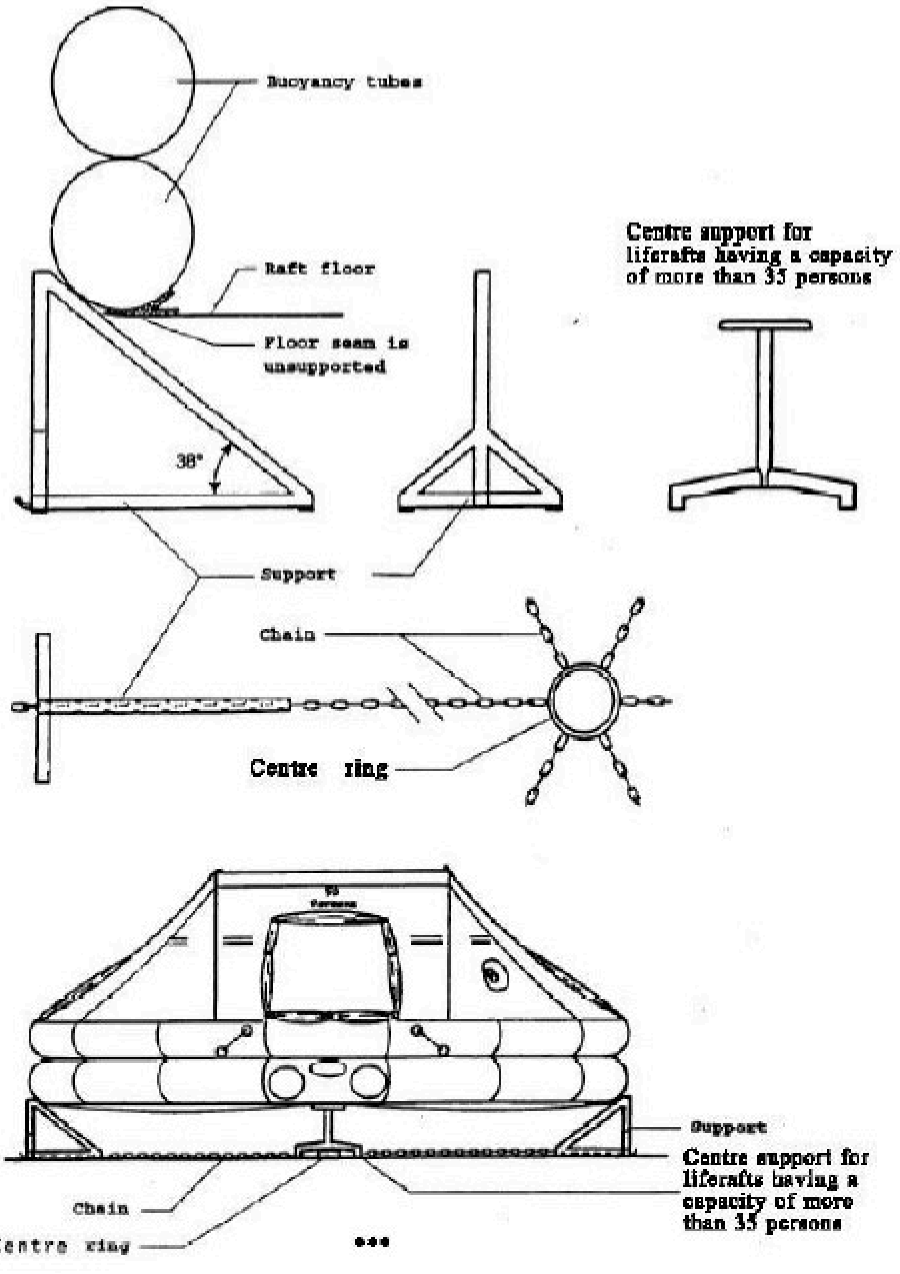
*may be at end of the sixth year if so recommended by the Manufacturer

**Unless the cylinder is replaced and then at five year intervals

Appendix 4

Appendix 4

Guidelines for floor seam test supports (Ref. paragraph 5.8)



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